



Stochastic Processes

Lecture 23

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Conditional Expectation

Fix a probability space $(\Omega, \mathcal{F}_0, \mathbb{P})$.

Definition (Conditional Expectation)

Let $\mathcal{F} \subset \mathcal{F}_0$ be a sub- σ -algebra of $\mathcal{F}_0$.

Let $X : \Omega \rightarrow \mathbb{R}$ be measurable with respect to $\mathcal{F}_0$ **with** $\mathbb{E}[|X|] < +\infty$.

The conditional expectation of X given $\mathcal{F}$, denoted $\mathbb{E}[X \mid \mathcal{F}]$, is **any random variable** Y satisfying:

1. Y is measurable with respect to $\mathcal{F}$.
2. For all $A \in \mathcal{F}$, we have

$$\mathbb{E}[X \mathbf{1}_A] = \int_A X \, d\mathbb{P} = \int_A Y \, d\mathbb{P} = \mathbb{E}[Y \mathbf{1}_A].$$

Two immediate questions:

- Given $\mathcal{F}$ and X which is measurable w.r.t. $\mathcal{F}$, does there exist at least one Y satisfying **1, 2** above?
- If there exists at least one Y , is it unique?

Existence of Conditional Expectation

- Existence of conditional expectation comes from the Radon–Nikodym theorem
 - Suppose X is a non-negative RV w.r.t. $\mathcal{F}_0$, and $\mathcal{F} \subset \mathcal{F}_0$
 - Define $\nu : \mathcal{F} \rightarrow [0, +\infty)$ as

$$\nu(A) = \int_A X \, d\mathbb{P}, \quad A \in \mathcal{F}.$$

- Then, ν is a measure, and $\nu \ll \mathbb{P}$

By the Radon–Nikodym theorem, there exists a non-negative, $\mathcal{F}$ -measurable function $Y : \Omega \rightarrow [0, +\infty)$ such that

$$\nu(A) = \int_A Y \, d\mathbb{P}, \quad A \in \mathcal{F}.$$

The RV Y is then a valid choice for $\mathbb{E}[X \mid \mathcal{F}]$

Integrability of Conditional Expectation

Lemma (Integrability of Conditional Expectation)

Suppose X is a RV w.r.t. $\mathcal{F}$ satisfying $\mathbb{E}[|X|] < +\infty$, and Y is any random variable such that:

1. Y is measurable with respect to $\mathcal{F}$.
2. For all $A \in \mathcal{F}$,

$$\mathbb{E}[X \mathbf{1}_A] = \int_A X d\mathbb{P} = \int_A Y d\mathbb{P} = \mathbb{E}[Y \mathbf{1}_A].$$

Then, $\mathbb{E}[|Y|] < +\infty$.

Proof of Lemma 1:

- Let $A_* := \{Y > 0\}$. Clearly, $A_* \in \mathcal{F}$. We then have

$$\begin{aligned} \mathbb{E}[Y_+] &= \mathbb{E}[Y \mathbf{1}_{A_*}] = \mathbb{E}[X \mathbf{1}_{A_*}] \leq \mathbb{E}[|X| \mathbf{1}_{A_*}] \leq \mathbb{E}[|X|] < +\infty, \\ \mathbb{E}[Y_-] &= \mathbb{E}[-Y \mathbf{1}_{A_*^c}] = \mathbb{E}[-X \mathbf{1}_{A_*^c}] \leq \mathbb{E}[|X| \mathbf{1}_{A_*^c}] \leq \mathbb{E}[|X|] < +\infty, \end{aligned}$$

Uniqueness of Conditional Expectation

- Suppose Y_1 and Y_2 are two choices for $\mathbb{E}[X \mid \mathcal{F}]$
- Then, by definition, we have

$$\int_A X \, d\mathbb{P} = \int_A Y_1 \, d\mathbb{P} = \int_A Y_2 \, d\mathbb{P} \quad \forall A \in \mathcal{F}.$$

- From Lemma 1, we know that

$$\mathbb{E}[Y_1] < +\infty, \quad \mathbb{E}[Y_2] < +\infty$$

- Fix an arbitrary $\varepsilon > 0$, and let

$$A_\varepsilon := \{Y_1 - Y_2 \geq \varepsilon\}.$$

- Noting that $A_\varepsilon \in \mathcal{F}$, we have

$$0 = \int_{A_\varepsilon} (X - X) \, d\mathbb{P} = \int_{A_\varepsilon} (Y_1 - Y_2) \, d\mathbb{P} \geq \varepsilon \mathbb{P}(A_\varepsilon),$$

thereby implying that $\mathbb{P}(A_\varepsilon) = 0$

- Because the choice of $\varepsilon > 0$ is arbitrary, we get

$$\mathbb{P}\left(\{Y_1 - Y_2 \geq \varepsilon\}\right) = 0 \quad \forall \varepsilon > 0.$$

Uniqueness of Conditional Expectation

- This implies that

$$\mathbb{P}\left(\{Y_1 - Y_2 > 0\}\right) = 0 \quad \Longleftrightarrow \quad \mathbb{P}\left(\{Y_1 \leq Y_2\}\right) = 1.$$

- Reversing the roles of Y_1 and Y_2 in the previous set of arguments, we can similarly show that

$$\mathbb{P}\left(\{Y_2 \leq Y_1\}\right) = 1$$

- It then follows that

$$\mathbb{P}\left(\{Y_1 = Y_2\}\right) = 1.$$

Takeaway

Conditional expectations are only unique almost surely.

Examples

Given

$$(\Omega, \mathcal{F}_0, \mathbb{P}),$$

X RV with respect to $\mathcal{F}_0$,

$$\mathcal{F} \subset \mathcal{F}_0.$$

- Suppose X is $\mathcal{F}$ -measurable. Then,

$$\mathbb{E}[X \mid \mathcal{F}] = X.$$

- Suppose X is independent of $\mathcal{F}$, i.e., $\sigma(X) \perp \mathcal{F}$. Then,

$$\mathbb{E}[X \mid \mathcal{F}] = \mathbb{E}[X].$$

- Suppose $\mathcal{F} = \{\emptyset, \Omega, B, B^c\}$ for some fixed $B \subset \Omega$ with $\mathbb{P}(B) \in (0, 1)$. Then,

$$\mathbb{E}[X \mid \mathcal{F}] = \left(\frac{\mathbb{E}[X \mathbf{1}_B]}{\mathbb{P}(B)} \right) \mathbf{1}_B + \left(\frac{\mathbb{E}[X \mathbf{1}_{B^c}]}{\mathbb{P}(B^c)} \right) \mathbf{1}_{B^c}.$$

- By convention,

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)].$$

Examples

Given

$$(\Omega, \mathcal{F}_0, \mathbb{P}),$$

X RV with respect to $\mathcal{F}_0$,

$$\mathcal{F} \subset \mathcal{F}_0.$$

- If $\mathbb{P}(B) = 0$ in the previous example, then

$$\mathbb{E}[X \mid \mathcal{F}] = \left(\mathbb{E}[X] \right) \mathbf{1}_{B^c} + c \mathbf{1}_B,$$

where $c \in \mathbb{R}$ is any arbitrary constant

- More generally, if $\mathcal{F} = \sigma(B_1, B_2, \dots)$ for some partition $\{B_n\}_{n \in \mathbb{N}}$ of Ω with $\mathbb{P}(B_n) \in (0, 1)$ for all n , then

$$\mathbb{E}[X \mid \mathcal{F}] = \sum_{n \in \mathbb{N}} \left(\frac{\mathbb{E}[X \mathbf{1}_{B_n}]}{\mathbb{P}(B_n)} \right) \mathbf{1}_{B_n}.$$

- **Law of iterated expectations:**

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}]] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mathbf{1}_\Omega] = \mathbb{E}[X \mathbf{1}_\Omega] = \mathbb{E}[X].$$

Examples

- Suppose X and Y are $\mathcal{F}_0$ -measurable, jointly continuous, and have a joint PDF $f = f(x, y)$
- Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be measurable with $\mathbb{E}[|g(X)|] < +\infty$
- **How do we compute $\mathbb{E}[g(X) \mid Y]$?**

- Given $A \in \sigma(Y)$, writing $A = \{Y \in B\}$ where $B \in \mathcal{B}(\mathbb{R})$, we have

$$\begin{aligned} \mathbb{E}[g(X) \mathbf{1}_A] &= \mathbb{E}[g(X) \mathbf{1}_B(Y)] = \int_B \int_{\mathbb{R}} g(x) f(x, y) \, dx \, dy = \int_B \frac{\int_{\mathbb{R}} g(x) f(x, y) \, dx}{f_Y(y)} f_Y(y) \, dy \\ &= \int_B h(y) f_Y(y) \, dy = \mathbb{E}[h(Y) \mathbf{1}_B(Y)] = \mathbb{E}[h(Y) \mathbf{1}_A]. \end{aligned}$$

- From the above, it follows that

$$\mathbb{E}[g(X) \mid Y] = h(Y), \quad h(y) = \begin{cases} \int_{\mathbb{R}} g(x) \frac{f(x, y)}{f_Y(y)} \, dx, & f_Y(y) > 0, \\ \text{any arbitrary value,} & f_Y(y) = 0. \end{cases}$$

Properties of Conditional Expectations – 1

Lemma (Properties of Conditional Expectation)

Let X and Y be $\mathcal{F}_0$ -measurable with finite expectations, and let $\mathcal{F} \subset \mathcal{F}_0$.

1. **Linearity:** $\mathbb{E}[aX + bY \mid \mathcal{F}] = a \mathbb{E}[X \mid \mathcal{F}] + b \mathbb{E}[Y \mid \mathcal{F}]$ for all $a, b \in \mathbb{R}$.

2. **Monotonicity:** If $\mathbb{P}(\{X \leq Y\}) = 1$, then

$$\mathbb{E}[X \mid \mathcal{F}] \leq \mathbb{E}[Y \mid \mathcal{F}].$$

3. **Monotone convergence:**

If $\{X_n\}_{n \in \mathbb{N}}$ is a sequence of non-negative random variables with finite mean, and $X_n \xrightarrow{\text{a.s.}} X$ where $\mathbb{E}[X] < +\infty$, then

$$\mathbb{E}[X_n \mid \mathcal{F}] \xrightarrow{\text{a.s.}} \mathbb{E}[X \mid \mathcal{F}].$$

4. **Conditional Jensen's inequality:**

If φ is a convex function, and X is a RV with $\mathbb{E}[|X|] < +\infty$ and $\mathbb{E}[\varphi(X)] < +\infty$, then

$$\mathbb{E}[\varphi(X) \mid \mathcal{F}] \geq \varphi(\mathbb{E}[X \mid \mathcal{F}]).$$

Properties of Conditional Expectations

Lemma (Tower Property)

If $\mathcal{G}_1 \subset \mathcal{G}_2$, then

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}_2] | \mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X | \mathcal{G}_1] | \mathcal{G}_2] = \mathbb{E}[X | \mathcal{G}_1].$$

Thus, the smaller σ -algebra always wins in conditioning.

Proof of Lemma 3:

- The second equality is obvious: $\mathbb{E}[X | \mathcal{G}_1]$ is measurable with respect to $\mathcal{G}_1$ and hence with respect to $\mathcal{G}_2$
- To show that the first and third quantities are the same, we note that for any $A \in \mathcal{G}_1 \subset \mathcal{G}_2$,

$$\mathbb{E}[\mathbb{E}[X | \mathcal{G}_2] \mathbf{1}_A] = \mathbb{E}[X \mathbf{1}_A] = \mathbb{E}[\mathbb{E}[X | \mathcal{G}_1] \mathbf{1}_A].$$

- The first equality above follows by noting that $A \in \mathcal{G}_2$ also, as $\mathcal{G}_1 \subset \mathcal{G}_2$

Martingales

Martingale

Fix a probability space $(\Omega, \mathcal{F}_0, \mathbb{P})$.

Definition (Martingale)

Let $\mathcal{G} = \{\mathcal{G}_n\}_{n \in \mathbb{N}}$ be any filtration, and let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a process such that

1. $\mathbb{E}[|X_n|] < +\infty$ for all n .
2. X_n is $\mathcal{G}_n$ -measurable for each n .

Then, $\mathbf{X}$ is called a **martingale with respect to the filtration $\mathcal{G}$** if

$$\mathbb{E}[X_{n+1} \mid \mathcal{G}_n] = X_n \quad \forall n \in \mathbb{N}.$$

- If the condition

$$\mathbb{E}[X_{n+1} \mid \mathcal{G}_n] \geq X_n \quad \forall n \in \mathbb{N}$$

holds, then $\mathbf{X}$ is called a **submartingale**

- If the condition

$$\mathbb{E}[X_{n+1} \mid \mathcal{G}_n] \leq X_n \quad \forall n \in \mathbb{N}$$

holds, then $\mathbf{X}$ is called a **supermartingale**